

Application of Trigonometry in Solving Real Life Problems

Introduction

Trigonometry, a branch of mathematics dealing with the relationships between angles and sides of triangles, plays a vital role in various real-life applications. From architecture and engineering to navigation and astronomy, trigonometric principles are crucial in solving practical problems. This project explores the application of trigonometry in real-life contexts and seeks to demonstrate how understanding trigonometry can enhance problem-solving abilities.

Stage 1: Problem Identification

Problem Statement

Many students struggle to see the relevance of trigonometry beyond the classroom. This disconnect often results in poor comprehension, low motivation, and subpar performance in trigonometry topics. Yet, trigonometry is essential in solving real-world problems such as measuring inaccessible heights, designing buildings, and calculating distances in navigation. There is a need to bridge the gap between theory and real-life applications in the teaching of trigonometry.

Initial Solution Statement

To address this problem, this project proposes using practical examples and fieldwork-based activities that demonstrate real-world uses of trigonometry. Examples include using trigonometry to measure the height of trees or buildings, estimating distances using angle measurements, and modeling construction designs. Such hands-on approaches can increase student engagement and improve conceptual understanding.

Stage 2: Investigation of Related Ideas

Literature Review Report

Research shows that:

- Trigonometry is fundamental in architecture, astronomy, and surveying.
 - Students benefit from contextual learning and problem-based approaches.
 - Curriculum models that integrate real-life applications improve learning outcomes.
- Various studies emphasize the effectiveness of experiential learning in mathematics.

Data Collection and Analysis Report

Data collection methods:

- Surveys of students to assess understanding and attitudes.
- Interviews with mathematics teachers.
- Observations of trigonometry lessons.

Findings:

- Most students struggle to connect trigonometry with real life.
- Teachers want to incorporate practical activities but lack resources.
- Observed lessons were predominantly theoretical.

Strengths:

- Mixed methods gave a broad view.
- Teacher interviews offered expert insight.

Weaknesses:

- Limited observation time.
- Surveys were confined to one school.

Stage 3: Generation of Ideas

Idea Generation Report

Generated ideas included:

1. Using clinometers to measure building heights.
2. Incorporating navigation apps to demonstrate angle and distance estimation.
3. Modeling triangular roof structures using trigonometric principles.
4. Creating project-based tasks like designing a bridge or ramp.
5. Using satellite imagery and maps to calculate distances.

Experimentation and Testing Report

Three main ideas were tested:

- Group A used clinometers for height measurements.
- Group B calculated distances using angle measurements and maps.
- Group C designed and modeled triangular roof structures.

Outcomes:

- All groups reported increased engagement.
- Group A understood angle of elevation concepts better.
- Group B improved accuracy in applying trigonometric ratios.

- Group C demonstrated creative application in design and modeling.

Stage 4: Development of Ideas

Idea Evaluation and Selection Report

Ideas were evaluated based on feasibility, cost, and learning impact:

- Clinometer use was low-cost and effective.
- Map-based exercises used accessible materials.
- Modeling required more preparation but led to deeper understanding.

Final choice: Combine all three ideas into a hands-on trigonometry application project.

Implementation Plan

1. Begin with a classroom introduction to trigonometric ratios.
2. Divide students into groups for specific activities:
 - Group A: Measure school building heights using clinometers.
 - Group B: Estimate distances on a map using angle and side calculations.
 - Group C: Construct and model roof trusses using cardboard and measured angles.
3. Present group findings.
4. Conduct reflection sessions and quizzes to assess understanding.

Stage 5: Presentation of Results

Application of Trigonometry

1. Architecture and Construction: Architects use trigonometry to calculate slopes of roofs, heights of structures, and determine loads and forces on buildings.
2. Navigation (GPS and Ships): GPS systems use triangulation (a subset of trigonometry) to determine a receiver's exact location based on distances to multiple satellites. It is

also crucial for sea navigation to define coordinates.

3. Surveying and Engineering: Surveyors use trigonometry to determine the height of mountains, slope of land, and boundary lines. It's used in building infrastructure like bridges and tunnels.

4. Astronomy: Astronomers use trigonometric methods to measure the distances to nearby stars and map the positions of celestial objects.

5. Medical Imaging: Technologies like ultrasound and CT scans rely on trigonometric functions to generate images of the human body.

6. Physics and Engineering: It is used to analyze waveforms (sound, light) and study periodic motion.

7. Criminology and Forensics: Trigonometry helps determine the trajectory of projectiles (like bullets) and analyzing skid marks to determine the speed of vehicles in accident investigations.

8. Computer Graphics/Game Development: Trigonometry is heavily used in coding to create 3D graphics, rotate objects, and determine perspectives in gaming.

Evaluation Report

Students demonstrated improved comprehension of trigonometric applications. Engagement increased due to hands-on activities, and test results reflected a positive shift.

Recommendations Report

1. Use real-world tasks to teach trigonometry.
2. Equip classrooms with basic tools like clinometers.
3. Encourage interdisciplinary projects with geography or design subjects.
4. Train teachers on implementing active learning strategies.

Conclusion

This project highlighted the importance of applying trigonometry in real-life scenarios. Through carefully planned activities, students bridged the gap between abstract concepts and practical understanding, leading to deeper learning and problem-solving abilities.